SIGMIDI FOOT CONTROLLER

v0.1 / KICAD ELECTRICAL SCHEMATIC / external board pins only

Continuous wires connect every fitted terminal. Dots = junctions; crossings without dots = no connection. Component placement is schematic.

Nano and LCD pins keep physical order. Nano: JP1 left 1–15, JP2 right 15–1, USB-C up. Outlines are schematic.

3V3 / Nano logic supply

+5V LED / MIDI

J1 5.5 x 2.1mm

9V DC / center-negative

SLEEVE (+)

CENTER (–)

C1 47uF / 35V

U3 S13V10F5

Pololu #4083 / 1178565

VIN GND 5V OUT

J2 ACJS-MHDC / passive pedals

TOP / pedal 1

11 RING

12 TIP

10 SLEEVE

BOTTOM / pedal 2

5 RING

6 TIP

4 SLEEVE

1/2/3/7/8/9: NC; pad 13: mechanical / NC

R20 5.1k

R30 5.1k

USB-C / MIDI

U1 Arduino Nano ESP32

Component-side top view / USB-C up

JP1:1 D13

JP1:2 / 3V3 3V3

JP1:3 B0

JP1:4 / GPIO1 A0

JP1:5 / GPIO2 A1

JP1:6 A2

JP1:7 / GPIO4 A3

JP1:8 / GPIO11 A4

JP1:9 A5

JP1:10 A6

JP1:11 A7

JP1:12 / VBUS VBUS / VUSB

JP1:13 B1

JP1:14 / GND GND

JP1:15 / VIN VIN

JP2:15 / GPIO47 D12

JP2:14 / GPIO38 D11

JP2:13 / GPIO21 D10

JP2:12 / GPIO18 D9

JP2:11 / GPIO17 D8

JP2:10 / GPIO10 D7

JP2:9 / GPIO9 D6

JP2:8 / GPIO8 D5

JP2:7 / GPIO7 D4

JP2:6 / GPIO6 D3

JP2:5 / GPIO5 D2

JP2:4 / GND GND

JP2:3 RST

JP2:2 / GPIO44 D0 / RX0

JP2:1 D1 / TX0

All printed labels retained; unused pads left unwired.

U2 Waveshare 24382

1 BL

2 RST

3 D/C

4 CS

5 SCL / CLK

6 SDA / DIN

7 GND

8 VCC

P2 / GH1.25 / SPI / pins 1–8

MIDI OUT / 2 x 3.5 mm TRS

D1 1N5819

D2 1N5819

MIDI_5V = converter +5V OR USB VBUS (D1/D2 1N5819)

U4 74HCT125 at MIDI_5V: 1A/2A from D0 / GPIO44

UART1 TX, 31.250 baud; R52 holds TX idle high

Unused 3A/4A and all ~OE to GND; 3Y/4Y open

C6 100nF across U4 pins 14/7

TRS Type A: T = sink (DIN 5), R = source (DIN 4)

S = shield / GND (DIN 2)

R52 10k

R53 220R

R54 220R

R55 220R

R56 220R

J3

MIDI OUT A

J4

MIDI OUT B

C6 100nF

U4 74HCT125

USB VBUS sense / 3.0V at 5V

R50 100k

R51 150k

SW11 ENCODER

A

20 detents + click

B

Internal pull-ups

SW1

No external R/C

C

SW2

Pololu 2535 / optional LEDs

LED1 / B

LED2 / A

LED3 / 2

LED4 / 1

LED5 / C

LED6 / 3

LED7 / D

LED8 / 4

LED9 / E

LED10 / 5

2535

DIN 5V GND DOUT

NC

R21 10k

R31 10k

MODULAR FOOTSWITCH LADDER

R7 10k

R8 20k

C5 100nF

R9 1k

R10 1k

R11 1k

R12 1k

R13 1k

R14 1k

R15 1k

R16 1k

R17 1k

R18 1k

SW1 B T1 / LED1

SW2 A T2 / LED2

SW3 2 T3 / LED3

SW4 1 T4 / LED4

SW5 C T5 / LED5

SW6 3 T6 / LED6

SW7 D T7 / LED7

SW8 4 T8 / LED8

SW9 E T9 / LED9

SW10 5 T10 / LED10

COMMON GND

PASSIVE PEDALS

Pot: outer legs RING / SLEEVE; wiper TIP.
Variable resistor: TIP to SLEEVE; RING unused.
0–10k per input. Bias HIGH in resistor mode;
high impedance in pot / disabled / boot states.

Netlist, BOM and pin references: design/electrical/. All 57 components and 49 nets checked against the connected drawing.

ASSEMBLY

Set 4 / 6 / 8 / 10 switches: calibrate installed taps.
SW11 contact pin arrangement is functional.
Native USB-C carries USB MIDI and ROM flashing.
USB-only supply does not power the external LED rail.

LED PACKAGE

Lens left, leads right, flat side down.
Top to bottom: DIN / 5V / GND / DOUT.
Only LEDs and wires; direct D7 / GPIO10 data.
Check 3.3V data acceptance on the actual LED batch.

Sheet: /
File: sigmidi-foot-controller.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.1 | Id: 1/1